

Full Q&A: Neuralink Patient Noland Arbaugh on His Injury, Selection, and Newfound Superpower

Noland Arbaugh shares his experience with Neuralink's brain-computer interface, and how it could unlock new possibilities for human enhancement and interaction with technology.



ALEX KANTROWITZ
DEC 19, 2024 · PAID



30



1

Share



Earlier this year, Noland Arbaugh allowed Elon Musk's Neuralink to drill a hole in his skull, insert more than one thousand electrodes into his motor cortex, and translate his brain signals into cursor movement and clicks. It's a remarkable story, and Big Technology [covered it in detail](#) yesterday.

It's really worth hearing it in Arbaugh's own words, where his optimism, sense of adventure, and true character come out. So today, I'm publishing the full transcript of the interview. You can read it below, edited lightly for length and clarity, or listen to our episode [Apple Podcasts](#), [Spotify](#), or [your app of choice](#).

Alex Kantrowitz: Noland, thank you for having me here at your home.

Noland Arbaugh: Absolutely, anytime.

You're the first Neuralink patient. That means you have a device on the back of your head that has about sixty threads that have been implanted into your brain. And you can control computers with your mind via the signals coming through the device. Is that the gist of it?

So, the implant is on the top of my head in the motor cortex on the left side of my brain. It's kind of like right on top, just off center to the left. There are 64 threads with 16 electrodes on each thread, totaling 1,024 electrodes in all, and they are implanted in the part of my motor cortex specifically correlated to right hand movement. Those electrodes, they pick up neuron spikes, neuron signals, and through an app on a computer right now called the Link App, it translates to cursor control.

So that mouse moving around your computer, it's not your eyes, it's not your fingers, it's legitimately your mind controlling it?

Yes.

Amazing.

Yeah, it's pretty cool.

We're gonna get into that more in a bit, but I first want to get a chance for our audience to learn about you and to get to know you. Let's just talk about the way that you ended up in this position, your injury.

When I was 22 years old, it was right after my senior year at Texas A&M, I was working at a summer camp in the pocono mountains called Island Lake Camp. It was our first day off, and a group of us went to a man made lake in Binghamton, New York.. There is a beach that was built that goes into the water. For all of you who are wondering, it's a sectioned off, cordoned off part of the water that's been there for probably decades, so there's nothing in the water. People always say “Oh, maybe there was a piece of a dock in the water in the lake, or there was something floating around.” It's not the case, I didn't dive in and hit my head on the bottom of the lake or anything. I just ran into the water with a couple of guys.

We all ran in together. We saw some girls in the water that were our friends who weren't really getting themselves wet. They had glasses on and their hair was dry. So we thought it would be hilarious to run in and pick them up and dunk them in the water. Three of us ran in together, and I got hit in the side of the head at some point as we were getting into the water, and right on the left side of my head. My c4 and c5 popped out of place and back into place, two vertebrae, and I woke up face down on the water.

And then what happened?

So, I woke up and I realized I couldn't move. I tried to move, and nothing happened. So immediately I realized, “Okay, well, I'm paralyzed.” Held my breath, hoping that maybe someone would come rescue me. No one did. So I held my breath as long as I could. Maybe 15 seconds. It felt like forever. It might have been 10 or 15 seconds, and I couldn't hold my breath any longer, so I decided, “Might as well take a big drink and whatever happens, happens.” And then for the next hour, I was sort of in and out.

In and out of?

Of consciousness. Most of it was me just unconscious. I woke up when they pulled me out of the water on the beach. I woke up again when an ambulance came and I was talking to a paramedic. I woke up again when they were transferring me to a helicopter, and then woke up again in the hospital right before they took me into surgery with a girl I had brought from Texas A&M standing over me just bawling, and I was trying to comfort her. I was saying, “Look, everything's gonna be fine. Look, I'm all right.” Things like that. I cracked some jokes. I don't remember what I said, trying to get her to laugh and stuff.

And I had a nurse standing next to me kind of telling me what was going on and what they were about to do. She had called my mom, and I said, “Don't tell my mom, it'll just stress her out. Let me go through surgery, and then you guys can tell her what happened.” And that was it.

And then you woke up?

Yeah, I was out of it for a while. They had me on Fentanyl and some Ativan and stuff. I like to tell people it was awesome. I never felt better. I saw some crazy stuff, woke up randomly and just hallucinated crazy things in my room. So that was a lot of fun. I felt great. But then the next couple weeks were pretty rough. Lots of pain. Lots of nerve pain. Just getting used to not being able to move. It was not fun.

So what can you feel? And what can't you?

Since I'm a C4, C5, anything below my shoulders, maybe a couple inches below my shoulders, I can't feel or move. So no sensation, no movement. I did gain back a little bit of movement in my arm and my hand but just a tiny bit. Just to move left and right. It's not much, but that's about it. It's a little bit of a bicep flex. That's about it. There's no feeling down there, no movement.

Is it numb or just non existent?

It just doesn't exist.

Being paralyzed is weird. I can see my body and I can feel myself trying to move, and then I just watch nothing happen. So it's odd. It's something you have to get used to. It feels like phantom limbs sometimes, where I feel, if I close my eyes and start trying to move my body, it seems like my body's moving, and then I open my eyes and nothing's happening. I have no sensation. So I have to be very careful about things that happen to my body, or what position I'm in for how long, because I can get pressure sores and things very easily, and I can hurt myself very easily. So it's just a matter of taking care of my body. I think I've managed pretty well with the help of my family.

So when did Neuralink come into the picture for you?

A little over a year ago. My buddy called me up just randomly drunk on a Wednesday at 11am and he said, “Hey, you want to get a chip in your brain?” I said, “Sure. Why not? I got nothing going on.”

You legitimately agreed to it on that first call?

Yes.

Your friend is persuasive. Or, you're daring.

Yeah. So my buddy is really big into the Elon sphere. He got drunk and was Googling SpaceX one day, and the human trials popped up on his computer, and immediately he thought of me. He worked in a spinal cord lab after college. He's a biology major, and I think part of it was because of me and my injury that he worked in that lab. So he's always thinking about me. He's always been a huge help to me and my family. So he called me up and said, “Let's do this.” And I said, “Yeah, sure. Why not?” He ran me through what Neuralink was, and I said, “That sounds pretty cool.”

I love that your friend calls you and says, “Elon Musk wants to put a chip in your brain.” And you said, “Yep. Yeah.”

The dude's done so many amazing things. My thought had always been that he's not going to put this in a human until it's ready. I feel like he's very pro-humanity, and he's not just gonna be shooting from the hip with this stuff and hurting people left and right just for progress. I never thought he would do that.

So when I heard that human trials had opened up through my buddy Bane, I knew that he was ready, and I had complete faith in the company and what they were doing. It sounded really cool. So he applied with me over the phone. It's kind of an interesting juxtaposition, because I couldn't apply on my own before, like it would have probably taken me an hour to apply online, if it worked at all. Now, I could probably do it in a few minutes with the Neuralink. It's pretty interesting. So he applied with me over the phone. He would ask questions and I would answer, telling him what to put. The questions were things like “What do you want from Neuralink?” or, “What do you want to get from this?” I would answer “An Iron Man suit.”

Just kind of joking around?

Yeah, I thought that it was the longest of shots, because there was nothing really spectacular about me. I feel like I didn't do anything with my accident, like I didn't accomplish anything in seven years. I just thought there was no way that they would pick someone like me. So I was kind of joking around on my application, not taking it very seriously.

Then you got a phone call?

Then I got an email like a day later. They said, “Oh, we're inviting you to apply.” They sent me a link with 20 different time slots. And my first thought was, “I know I'm not very memorable, so the least I can do is select the very first time slot on everything they send me,” and at least they will remember me for being the first person to be interviewed and stuff.

And so what was the interview like?

The initial one, the very, very first one was basically just a screening. They asked things like “Hey, what's your name? How old are you? What’s your date of birth? Are you paralyzed?” As in, “Are you *actually* paralyzed?” Just to make sure that I wasn't some random person applying. I told them I'm just a normal guy and I just really want the Neuralink. So that was the first interview.

The next interview started going into things like my medical history. More in depth into my injury and how I lived now and where I lived. Things like that. And then it progressively kept becoming more detailed. I had calls with surgeons, I had calls with psychiatrists to make sure I wasn't a crazy person. And then eventually, consent forms that said things like “Okay, well, if you do the study, this is what you're consenting to,” and things like that. And that all culminated with an in person day-of testing at a hospital where they did body scans, brain scans, blood test, urine test, a real psych evaluation, a memory evaluation, all sorts of things. Eight hours of testing up in Phoenix at Barrow Neurological Institute, which went pretty well.

And at this point, is it dawning on you that, “Oh, there's a real chance that I might get brain surgery to implant a digital device in there?”

I kept my expectations extremely level through this whole process. They told me constantly that if anything about my injury or my medical history didn't line up with exactly what they were looking for with their criteria, then they would say, “Okay, thank you, but we're going to go in a different direction.” They were always telling me that, in every interview saying, “Okay, this is going to be a two hour interview. If at any point what you say doesn't line up with what we want, then we'll say sorry.” I said, “Yeah, that's fine.” So I kept my expectations extremely level. Everyone around me didn't. My buddy called every day. He would say things like, “Wow– you made it through another one! You got another interview! You got another interview!”

So it was starting to feel real?

Yeah. I think the day I went up and got all the testing in Phoenix was when it all became very, very real to me.

So they said, “Okay, you're in.” And then they gave you some options about when exactly you might be able to do this?

So after my day of testing, it was still another month before they told me I was even selected as one of the participants. And then another month after that before they told me I was the first.

I don't know if I'd want to be the early adopter on this. I’d say “Let somebody else get it in their brain and see if it works.” What made you say, “Okay, let's make this work?”

A couple things. It's always cool to be the first at anything. I mean, I think that's pretty obvious. I think anyone given that opportunity would probably do it. And the next was, I knew that I would feel absolutely terrible and I would never forgive myself if I passed it up and someone else was hurt or injured or had to go through something terrible, because I decided to pass it up. I knew that I wouldn't ever want to do that, so I decided that if they gave me the opportunity, I would be number one. And they did,

What was surgery like?

That’s a great question. Who knows what they did? I was asleep. They put me under anesthesia. The surgery was supposed to last between three to six hours. It lasted under two. Everything went so perfectly. You can look up stuff about the needles and the cartridge for the R1 robot, which is the surgical robot.

It actually took over from the human surgeon to insert the electrodes into your brain?

Yep, we call it “Tiny Dancer.” The surgeon made the incision, drilled the hole in my skull, and then let the surgical robot, Tiny Dancer, take over. They had taken brain scans when I had done all of my testing, and they had mapped out where each thread was going to go, and they said that when they took off the piece of my skull, they were just hoping and praying that everything looked the same, that everything lined up, and they said it was absolutely perfect that they didn't have to make any adjustments, that Tiny Dancer went straight to work. They brought about 20 needles to replace the needles, because they didn't know if it would be able to survive the whole surgery, or if it would break or anything of that sort. One needle did it. So it was incredible. They said that everything went just about as perfectly as it possibly could have, and they closed me up in an hour and fifty-four minutes or something.

So they take a piece of your skull out, if I've read correctly, the robot puts these 64 threads into your motor cortex, and each of them has a set of electrodes on it. So you have one thousand-plus electrodes that are sensors that read the firing neurons in your brain, and then they effectively plug it up with the device, and then put the skin back over?

Yes, each of the 16 electrodes on the 64 threads is placed near neurons in my brain. They tried to get the electrodes near a certain depth in my brain, which is where all of the movements occur, really. So they put the electrodes up and down the thread just to make sure that at least one or a couple of the electrodes were close to the depth that they needed it to be, and they were picking up neurons firing in my brain that control motor movement, specifically to my right hand. All those signals are picked up in some way, and it's sent back to the Link app, and you can do some pretty cool things with them.

So you wake up from surgery and the first thing you do is play a prank on your mother?

Yeah, this is everyone's favorite part of the story. I think it's one of my favorite parts. So my buddy was staying with us for a couple weeks before my surgery to help out, and we were just joking around, thinking it would be so funny to do something out of surgery. Like, come out of surgery and wake up from anesthesia and just play a little prank on people. Bane didn't think I would do it. He said, “There's no way you do that.”

I woke up from surgery, and everyone's standing around me, and my mom looked at me. She asked, “Hey, honey. How are you doing? I just looked at her, and I said, “Who are you?” She freaked out.

When you came out of surgery, did anything feel different?

No, I've never felt different the whole time. I just feel like the same person, but I pulled a little prank on my mom. I thought it was hilarious. Can't believe I remembered it. Because I was so drugged up on anesthesia. I thought, “There's no way I remember to do this.” But it worked out okay.

Elon Musk came into the recovery room, into my hospital room, after I was recovering a bit. I think he came in probably a couple hours after my surgery. I was still a little groggy, and he came in saying, “Hey, good job. Finally made it here. It's good to meet you.” Things like, “How do you feel? How's it going?” Things of that nature. And I don't remember much, but what I do remember very vividly is how cool his bomber jacket was. It was a sweet bomber jacket, but it worked out okay. Got to meet him, kind of sort of, and then a little bit later the team came in and they watched [the Neuralink] kind of work for the first time.

When you say they watched the Neuralink work, talk a little bit about how your brain signals started showing up through the device.

So the first thing they did was come in and woke the device up, and when they heard the first “ding” for the first time of it waking up, everyone was so happy. There were just sighs of relief. Exclamations, like, “Oh my gosh.” and “It's amazing!” sort of thing.

Everyone was just so happy. And then they connected it to a tablet to show some brain signals in real time, and they held the brain signals in front of me, and I was looking at eight different channels. One of the my channels is eight different electrodes that were picking up neuron signals. And I got to see in real time my brain signals, my neuron spikes.

If you've seen the Neuralink symbol, the Neuralink logo, it's a neuron spike. Basically, it's what they look like. It's almost like a heartbeat monitor. It's just a little bit different, a little bit more erratic. And so they showed me eight different channels, and I still remember so vividly, the third box over on the top row, because it was four-by-four, and when I saw the signals, I thought, "Can I manipulate this at all?" So I started wiggling my fingers and my toes and trying to move. And I saw a very specific yellow spike when I moved my right index finger.

And this is doing it physically, or just thinking about it?

Physically, trying to move my finger. I'm just attempting to move my finger, and I saw this big yellow spike every time. And I said, "Oh, that's cool." And everyone was asking, "What?" as in, "What are you doing?"

I said, "Look at the third box on the top right now... And now... And now. I said, "That's my index finger." And everyone freaked out.

Because that meant the device could read what was going on in your brain and then eventually translate that into something on the computer?

Yes, in real time. It was incredibly effective, honestly, just a few hours after my surgery.

How do you go from having that experience of "I'm gonna try to move my finger and it's gonna show up in one of these eight channels" to then being able to move things around on a computer, just thinking about them?

It's calibration. When you calibrate anything, calibrate a mouse, calibrate a controller, a TV, you have to basically train it to do what you want it to do. So with a cursor, they will show a visualization, like an animation of a cursor moving, right, left, up, down, diagonally and stuff. And all I have to do is follow it with an attempted movement with my hand in some way, and I can reach for it. I can move my wrist in that direction. I can move my hand in that direction. I can use my whole arm. I can use my feet, or anything, just any body part. I use my right hand, because that's where the strongest signals come from. And you just follow the cursor. And then eventually, all of those times that you follow the cursor, all of those times that you've followed, kind of the White Rabbit, if you will, the Link app, the algorithm, the machine learning, is taking those brain signals and teaching this algorithm to associate those movements with the neurons that are firing when it's happening, and eventually you calibrate a cursor and you get control of it.

And so how long did it take you to start being able to control computers using that type of brain?

So for the first 10 days, they weren't allowed to charge my implant, because of safety regulations, they had to make sure that it was safe in my brain first, that it kind of stabilized, that the temperature range was okay. And so they weren't allowed to charge at all, because charging raises the temperature, and this is a safety study first. So they needed to make sure that everything was good. So every day, for the first 10 days, some Neuralink guys would come over to the VRBO that we were staying at first, near the hospital. They'd come over, just two of them, they would wake it up, and we would work for an hour tops. And they would say, "Okay, we used 11% that's it for today." And we would do different tasks. We would do what we call "body mapping," which is an animation of a hand moving different fingers in different ways, moving across the screen. Just all sorts of different movements that they wanted me to replicate, to follow.

When you're using a mouse, or when you're moving, you have to first think and then move. But the Neuralink can track your intent even before you even realize that it's happening.

There's a lot about the implant that I find absolutely fascinating. So first off, let's go through a couple terms. You have attempted movement and imagined movement. Attempted movement is literally me attempting to move my hand. Pretty self explanatory. If the cursor is going left, I might try to move my hand to the left. That's me physically attempting to move it. Even though nothing's happening here, all the signals still work in my brain. There's just a disconnect in my spinal cord so they don't get through, but the Neuralink is picking up those signals. So that's attempted movement. I'm physically trying to move my hand, my fingers, my legs, whatever it is, in whatever direction, in whatever motion, to it to a keystroke, map it to a directional control of a cursor, or something of that nature. So that's attempted movement.

And then we have imagined movement. Imagined movement is basically the same, except you don't physically try to move a body part. You just think about either moving the body part, or you just think of moving a cursor or doing an action that you want the cursor to perform. So you just think, "Cursor, move to the right, left, up, down." You think, "Cursor, left-click, cursor, right-click." And you do these things all just by thinking. And so there's a difference.

This has been BCI – "brain-computer interface" – terminology for a while, this is not something that we've come up with. So you take those things and eventually you get some really, really interesting outcomes with the algorithm and the implant learning your intention when you go to move. You don't think, "Move hand here." You just move your hand wherever you want it to go. You don't think, "Grab this." You just do it. And you grab before you physically move, the signal from your brain has to go all the way down and all the way back up, it shoots down to your hand, saying, "Move hand this way," all the way back up to your brain. And then the action happens.

The same goes for me as I'm trying to attempt to move, or I'm imagining moving. And so the implant, the algorithm, can anticipate what you're doing, just like your body is firing that signal before you even move. So sometimes when I want to say click or I want to move, the implant will understand my intent and move before I even decide to move.

That's crazy.

It is crazy. And so I think that the implant is much more capable than a lot of people understand. It really is fascinating.

I'm thinking about video games, and if you're playing video games, that's all coming through your hands, but if you're able to move things on your computer first, you could probably just destroy anybody in any game with this. It's a superpower.

Yeah, when I think of where this technology will go, it's pretty incredible. I don't believe it will be allowed in certain gaming platforms and things for Major League Gaming. I think they're going to have to separate it. Either everyone's going to have to have one, or you're going to have to understand that other people have one and just accept that maybe it's better, or you're better. Maybe assume, "Oh, the neuralink doesn't add any sort of advantage. So I'll play against it." I don't believe that's true. I think they're going to have to separate them to their own league, or ban them in leagues. Or everyone's gonna have to have them in order to be fair, because they really are, like you said, a superpower. When it comes to video games, your brain has to send the signal to your hands and come all the way back up in order for you to move the joysticks or click buttons on the controller. It doesn't have to do that with a Neuralink. It's all happening sort of instantaneously. Or pre-instantaneously, if it's learning your intent, in your mind, and it's sending that signal to a computer and back. I'm not sure what the latency is, but I have a feeling it's less than whatever you're doing with your hands with a controller. So it's going to be a few milliseconds are life or death in these sorts of games, right?

It's not there yet, but I know that when this technology really lands there's no doubt in my mind that people will want it in order to surpass what “regulars” are capable of.

I love that you're just referring to folks as “regulars.”

Yeah, it's amazing what you're able to do.

So let me just ask you this question. It's a simple question, but I'm curious to hear your answer. I mean, what does it feel like to not be able to move your hand, but to see that movement expressed on a screen?

What's interesting is, for seven years, seven and a half years, I was trying to move my hand and my body to create new neural pathways to hopefully regain some movement. I never knew if it was working, because there's no feedback loop. I try to move my hand, I try to move my feet, I try to move my legs, my body – in any way – but I don't know if it's working – if what I'm trying to do is actually being propagated in my brain. Really, I don't know if trying to move my hand, if the signals are actually firing in my brain. When I was able to see my neurons firing for the first time, it really made me kind of emotional thinking that when I move my index finger, there are still neurons firing in my brain. It's really just my spinal cord. And I think that should give a lot of people hope, honestly, because everything's still working. You've just gotta keep trying and keep forcing your body with brute force to relearn how to move.

Even though you don't have that sensation back, you do have it back in some way.

It's weird. It's a weird psychological leap, and sort-of physical leap. It's weird to be able to move without moving.

So you use the Neuralink app, and that allows you to control your cursor on the screen. So once you got past the point of saying, “Okay, I can do this,” what has that enabled you to do that you couldn't do before?

I mean, so much. It's little things that add up to really big things. Being able to, like I said earlier, fill out an application online. It's given me a lot of hope and purpose and future prospects. For example, I might be able to get a job, I might be able to go to school. We were just talking about doing video editing. If I want to, I can just interact with the computer at a much, much higher level, play video games. Just enjoy surfing the web without having to go through the headache of the Mac OS navigation and things.

It adds up to a lot being able to send a text in a few seconds, instead of a text taking 5, 10, 15 minutes sometimes to craft because of how bad dictation is. I mean, it's terrible. There's so much that I'm able to accomplish with the computer now, that I was never able to do before, or that would have taken way too long. If you think about if it takes five minutes to send a text message to one person, imagine trying to keep up a conversation with 10 people in a night and sending 50 to 100 texts to each person. It's just not possible with Mac OS dictation if everything takes that long. With the Neuralink, it's way better. I feel like I've reconnected with my friends, with my family, with the world in general. Being able to communicate with people online, through social media, I really have just blossomed into a social butterfly with all of this.

It's interesting to me, because the digital world is so expensive, and for you, being confined in the way that you are physically, it must have felt like a release to just be, all of a sudden, able to access and interact on the internet the way that you are.

It's very “Ready Player One.” It's getting close. I'm not virtually in a “world” yet, but I sort of am, in a way. I'm interacting with people through a computer with the Neuralink from my bed as a quadriplegic like any person could from wherever they are in their lives. And I think that's amazing. Sometimes I forget that I'm even paralyzed and that I'm restricted in the way that I am, because it's just so seamless.

I can't imagine what it's going to do for the “normals.” I've been calling them, when they get this device and are able to do what I'm able to do while also physically going

about their lives. People are going to have a lot of fun with this, and they're going to play with it and think, "I don't understand," because I don't understand a lot. We don't understand the brain, and I'm constantly learning. But people will be able to go about their lives and do the dishes while they're doing stuff with their Neuralink, and using their hands and playing with the Neuralink or doing whatever. They're going to be able to multitask at such a higher level, and the Neuralink right now is capable of doing a limited amount of tasks at once. But I don't know that there's an upper limit to this. You can map as many things as you want to different actions. With the Neuralink, you think of being able to use both hands and both feet at the same time. It's pretty hard. You have to be pretty coordinated. Now imagine both hands, both feet, all 10 fingers, all 10 toes, shoulders, arms, elbows, knees, like every body part that you can imagine mapping to something else, and then that's all attempted moving. If you break that down even farther to imagined movement, it's unlimited how much potential and how many different tasks you'll be able to combine and do at once with Neuralink. It's incredible. I don't know where that limit is, but I'm positive that it's nowhere near where we are now. I know that it's going to keep growing.

So you think “normals,” as you call them, are gonna want to use it?

Oh, for sure, absolutely.

What would some use cases be?

Anything that anyone does on a computer. If they want to be more productive, if they want to be better at video games, if they just want to enjoy using it. Elon's plan is integrating with AI and virtual reality. Basically every device that you can think of that uses Bluetooth, you'll be able to control with the Neuralink. I constantly tell people, you think things online are Alexa-compatible. I have an Alexa-compatible fan. I have an Alexa-compatible Dyson. I have Alexa-compatible lights and smart plugs, all of that you're going to be able to do with the Neuralink. Everything is going to be Neuralink-compatible one day. Everything is going to be so you don't have to say it exactly. You just think. And so one day, I think everything is going to be Neuralink-compatible. Companies will have to put that in their product in order for it to sell. If it's not Neuralink-compatible, well, why are we even using this?

Neuralink is about to start testing a robotic arm. What do you think about that? And is that something that you'd want to participate in, if given the chance?

100% I'd want to participate in it. I'm not sure that I'll get the chance. I'm not sure that Neuralink wants me to be a part of all of that and if they do, I'm more than happy to. But there's a chance that maybe the next participants will have better control. They'll be able to control the robot arm a little bit better. I have like much less, threads and electrodes in my brain because of the thread retraction that happened about a month after my surgery.

Can you talk about what happened there?

So basically, immediately after surgery, and up to a month and a half after surgery, threads were being pulled out of my brain, or just moving in my brain.

And when you say it pulled out, it's the brain that's actually moving and spitting them out?

The brain moved much more than what Neuralink anticipated, which kind of blew my mind. Honestly, like you would think that we're at a place in human history and medicine, that we would know how much the brain moves, but they didn't. We know so little about the brain, which shocked me.

They drill the hole in my skull – it's called a craniectomy – you drill a hole and you pry something back over the hole in the skull, and you leave it open. But when they're drilling these holes in these craniotomies, they are doing it to relieve pressure in the brain, and so the hole doesn't need to be very big. My skull hole was really big. Not

super big, but big enough to fit the quarter size implant, plus its base, in my skull. So, bigger than a quarter. A little square piece is mounted to my skull.

So when they cut the hole in my skull, they were able to see the brain on a bigger scale than they're used to. The brain pulsed with my heart, they said, at a rate of three times as much as they thought. They thought that the brain moved a millimeter when it pulsed, and mine was moving three millimeters every time it pulsed. So when they implanted the threads, they didn't anticipate that, and they also didn't anticipate a couple of other factors, which ultimately led to thread retraction in my brain.

So that's one thing that happened to me that's pretty terrible. It was awful. It was a terrible experience. A month in, they basically told me that you have thread retraction, and I was losing control of a cursor. I was not able to control the cursor anymore at that point, and so I thought I was just out of the study, but they figured it out. They figured it out on the software side. Instead of registering like single neuron spikes, they were registering groups of neurons.

And so it was really hard, though, when they told me it was the day that I was going to visit the Neuralink facility in Fremont, it was the morning before, and they told me I had the thread retraction. They wanted to be as transparent as possible with me, which they have been through this entire study. And I thought I was out of the study. I thought it was over. I got to play with it for a month and not even see the peak of the mountain. I just got a little bit of a taste. And it was really, really hard. I cried in my van on the way to the Neuralink facility. I said, "This sucks. It's awful. I wasn't expecting this at all." And eventually I just said, "I'm here to do my part. I'm here so that these things happen to me and not the people that come after me. I'm here to do whatever I can to help this product, to help Neuralink." And so that's what I did.

It was hard to get there. It's not easy. It's not like you just flip a switch. It's something that I have to work through in my mind and come to a decision. I can be sad about this, or I can do my part and just focus on my purpose and understand why I'm really here and what it's going to do for the future, and that makes things a lot easier. So yeah, the thread retraction happened. So I don't know if with my thread retraction, because I have they said 15% of the electrodes are still in place on my brain, plus or minus 10% so it could be as little as five,

but still working with maybe 5% working.

It's one of the greatest pieces of this story, I think, that is hardly ever told, is that I'm able to do what I'm able to do with a fraction of the electrodes that were originally intended. And that is unbelievable. They changed the way that they are registering the neurons – the neuron spike – and they knew about it. They knew about it with the monkeys, but they said that it wasn't as good, that they really shouldn't use that type of neuron spike detection. But then in humans, it worked out way better. And I asked them one day, "So why didn't you guys do this from the beginning?" And they just said, "We didn't know." Like, this is what the study is for. So with my limited electrodes in my brain. And I'm with the control that I have of the cursor and my clicks and things, I'm not sure they're as good as anyone that's going to come after. They might have better control, because they solved the issue with p2. There was no threat of traction in Alex.

Alex is the second patient?

Alex is the second patient.

So you've met the second patient?

I haven't.

Oh, is he not going public?

I don't think so. I haven't heard a peep from him. I've mentioned that it's fine if he wants to meet me, but if not, it's his prerogative.

And how many Neuralink patients are there at this point?

I don't know.

So it's not clear to me if I will get a robot arm. If I do, I have some really fun plans. If they do give it to me, I think it'll be really cool. I'm not sure that what I want to do is strictly legal, but we'll find out.

I’ve got to come back to you when this happens.

So, yeah, I think if they give me a robot arm, people are going to be terrified.

Wait. Are you going to shoot guns with a robot?

I don't want to say what I'm going to do, but let's just say it will be really cool if they give it to me. And I might, I might ask for forgiveness instead of permission. So I don't know. We'll see. I have no idea if they'll give me one.

So now I want to get into the weird stuff. First of all, the singularity. One of the things that Elon's talked about with this is that Neuralink is a way to potentially protect humans from AI, or make humans more aligned with AI. Now that you have the device basically implanted in you. What do you think about that idea?

This post is for paid subscribers

Subscribe

Already a paid subscriber? Sign in

← Previous

Next →